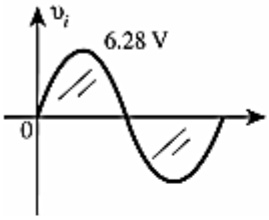
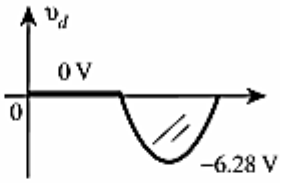
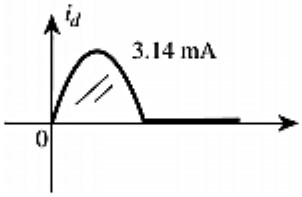
1

$$V_{dc} = 0.318\,V_m \Rightarrow V_m = \frac{V_{dc}}{0.318} = \frac{2\text{ V}}{0.318} = \mathbf{6.28\text{ V}}$$





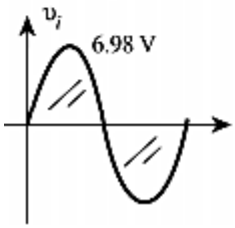
$$I_m = \frac{V_m}{R} = \frac{6.28\text{ V}}{2\text{ k}\Omega} = \mathbf{3.14\text{ mA}}$$

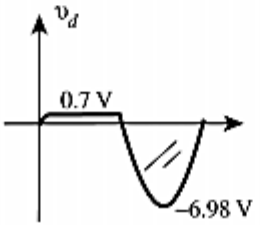


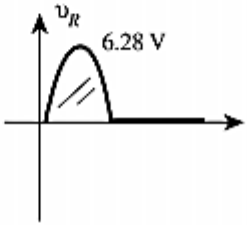
Using $V_{dc} \cong 0.318(V_m - V_T)$

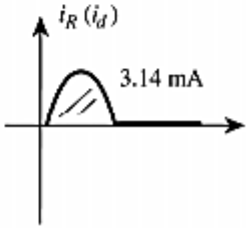
$2\text{ V} = 0.318(V_m - 0.7\text{ V})$

Solving: $V_m = \mathbf{6.98\text{ V}} \cong 10{:}1$ for $V_m{:}V_T$



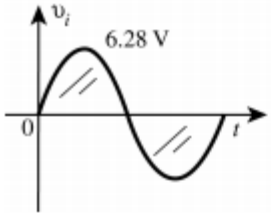


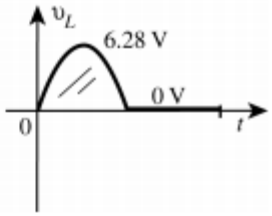




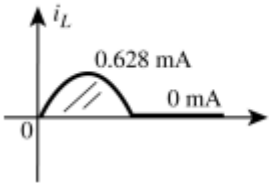
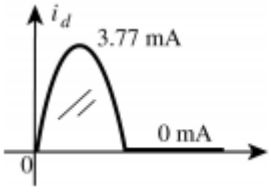
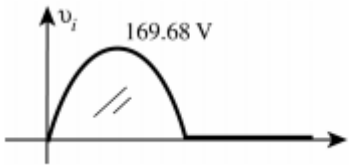
2

$$V_m = \frac{V_{dc}}{0.318} = \frac{2\text{ V}}{0.318} = \mathbf{6.28\text{ V}}$$





$$I_{L_{max}} = \frac{6.28\text{ V}}{10\text{ k}\Omega} = \mathbf{0.628\text{ mA}}$$

	 $I_{max}(2\ k\Omega) = \frac{6.28\ V}{2\ k\Omega} = \mathbf{3.14\ mA}$ $I_{D_{max}} = I_{L_{max}} + I_{max}(2\ k\Omega) = 0.678\ mA + 3.14\ mA = \mathbf{3.77\ mA}$ 
3	$V_m = \sqrt{2}\ (120\ V) = 169.68\ V$ $V_{dc} = 0.318V_m = 0.318(169.68\ V) = \mathbf{53.96\ V}$ 

4

Diode will conduct when $v_o = 0.7$ V; that is,

$$v_o = 0.7 \text{ V} = \frac{1\text{ k}\Omega(v_i)}{1\text{ k}\Omega + 1\text{ k}\Omega}$$

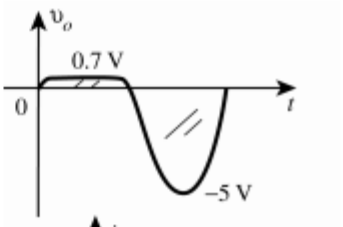
Solving: $v_i = \mathbf{1.4\ V}$

For $v_i \geq 1.4$ V Si diode is "on" and $v_o = \mathbf{0.7\ V}$.
For $v_i < 1.4$ V Si diode is open and level of v_o is determined by voltage divider rule:

$$v_o = \frac{1\text{ k}\Omega(v_i)}{1\text{ k}\Omega + 1\text{ k}\Omega} = 0.5\, v_i$$

For $v_i = -10$ V:

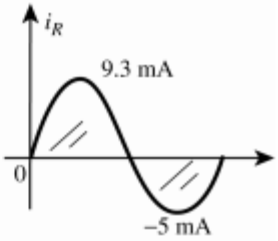
$$v_o = 0.5(-10\text{ V}) = \mathbf{-5\ V}$$



When $v_o = 0.7$ V, $v_{R_{max}} = v_{i_{max}} - 0.7$ V

$$= 10\text{ V} - 0.7\text{ V} = 9.3\text{ V}$$

$$I_{R_{max}} = \frac{9.3\text{ V}}{1\text{ k}\Omega} = 9.3\text{ mA}$$



$$I_{max}(\text{reverse}) = \frac{10\text{ V}}{1\text{ k}\Omega + 1\text{ k}\Omega} = \mathbf{0.5\ mA}$$

5

(a) $V_m = \sqrt{2}\,(120\text{ V}) = 169.7\text{ V}$

$$V_{L_m} = V_{i_m} - 2V_D$$
$$= 169.7\text{ V} - 2(0.7\text{ V}) = 169.7\text{ V} - 1.4\text{ V}$$
$$= 168.3\text{ V}$$
$$V_{dc} = 0.636(168.3\text{ V}) = \mathbf{107.04\ V}$$

(b) $\text{PIV} = V_m(\text{load}) + V_D = 168.3\text{ V} + 0.7\text{ V} = \mathbf{169\ V}$

(c) $$I_D(\text{max}) = \frac{V_{L_m}}{R_L} = \frac{168.3\text{ V}}{1\text{ k}\Omega} = \mathbf{168.3\ mA}$$

(d) $$P_{max} = V_D I_D = (0.7\text{ V})I_{max}$$
$$= (0.7\text{ V})(168.3\text{ mA})$$
$$= \mathbf{117.81\ mW}$$

6

(a) Center-tapped full-wave rectifier

(b) $V_{p(sec)} = (0.25)(1.414)120\ \text{V} = \mathbf{42.4\ V}$

(c) $\dfrac{V_{p(sec)}}{2} = \dfrac{42.4\ \text{V}}{2} = \mathbf{21.2\ V}$

(d) See Figure $V_{RL} = 21.2\ \text{V} - 0.7\ \text{V} = 20.5\ \text{V}$

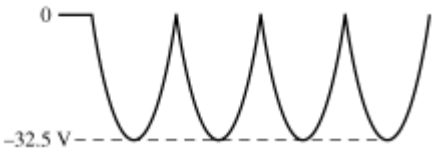


Figure 2-2

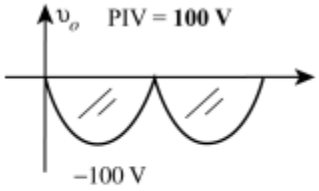
(e) $I_F = \dfrac{\dfrac{V_{p(sec)}}{2} - 0.7\ \text{V}}{R_L} = \dfrac{20.5\ \text{V}}{1.0\ \text{k}\Omega} = \mathbf{20.5\ mA}$

(f) PIV = 21.2 V + 20.5 V = **41.7 V**

7



8



$I_{max} = \dfrac{100\ \text{V}}{2.2\ \text{k}\Omega} = \mathbf{45.45\ mA}$